Dimitri Chrysafis

dimitri.chrysafis@gmail.com • 650-305-9037

DimitriChrysafis • dimitrichrysafis.github.io

Education

University of Wisconsin–Madison Graduating May 2027

Bachelor of Science in Computer Science **GPA: 4.0**

Technical Skills

ML Systems: LoRA/QLoRA fine-tuning, efficient quantization, adapter merging

GPU/Graphics Systems: real-time rendering, shader programming, mesh geometry, low-level GPU memory access, compute-shader kernels

---

Experience

SimpleX Chat **Open Source Contributor**

- Contributed to SimpleX Chat, an open-source private messaging platform, by improving multi-device sync logic in the client messaging pipeline
- Implemented sync-state reconciliation and connection lifecycle updates to reduce duplicate delivery edge cases across linked devices

---

Projects

Interactive 3D Fluid Simulator Using Material Point Method [Demo] [GitHub]

- Built browser-based physics engine implementing Material Point Method (MLS-MPM) with WebGPU compute shaders, simulating **400,000+ particles** in real-time with interactive camera controls and dynamic boundary animations
- Optimized particle-to-grid transfers through GPU parallelization and fixed-point arithmetic, achieving smooth performance for dam break scenarios and real-time particle injection

Real-Time Fractal Visualization Engine [Blog]

- Engineered real-time fractal visualization for Mandelbrot, Newton's, and Kleinian limit sets using GPU-accelerated WebGL fragment shaders optimized for deep zoom
- Achieved up to **60×** speedup through adaptive sampling, tile-based rendering, and optimized complex arithmetic

Volumetric Sphere Packing for 3D Meshes [GitHub] [Blog]

- Built mesh-based sphere packing system that fills arbitrary 3D models with non-overlapping tangent spheres using ray-mesh containment tests and nearest-surface distance queries
- Accelerated candidate search and clearance updates with PyTorch tensor operations, Open3D mesh processing, and live visualization for iterative packing of complex OBJ meshes

Tennis Match Prediction System

- Built ensemble classifier using CatBoost on **92,000 ATP matches (1982-2024)** obtained from publicly available historical datasets, engineering features including player ELO ratings, surface-specific performance metrics, head-to-head statistics, recent form indicators, and fatigue scores from tournament schedules to achieve **83.7% prediction accuracy**, outperforming baseline models by **12%**

Fourier Drawing Machine [GitHub] [Blog]

- Implemented a full Discrete Fourier Transform pipeline in Python that converts sampled drawing points into complex coefficients and reconstructs curves with rotating epicycles
- Built an animation renderer that generates frame-by-frame trajectory output with PIL/NumPy and exports smooth drawing visualizations from custom or image-derived point sets